

EE 364B Project Proposal: Inverse Optimal Power Flow via Differentiable Convex Optimization

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1 Background

Most wholesale electricity markets in the United States clear by solving, every few minutes, some variant of the *optimal power flow* (OPF) problem: a convex program that dispatches generation to meet load at minimum cost subject to network and engineering constraints [7]. The cost coefficients, line capacities, and ramping limits that parameterize this problem are, in principle, the defining knobs of the system; in practice they are either proprietary, approximated from heuristics, or drift in time with fuel prices, derates, and topology changes. Researchers who want to reason *downstream* about the grid — computing marginal emission factors (MEFs), pricing carbon, or stress-testing policy proposals — are therefore forced to work with models whose parameters are only loosely tied to reality [8, 14, 16].

At the same time, the field of *differentiable convex optimization* has matured to the point where this gap is addressable. Implicit differentiation of the KKT conditions [1–3, 5], packaged in `cvxpylayers`, lets us backpropagate through a convex program and fit its parameters to observed solutions. Donti et al. [11] applied this idea to OPF on a 14-bus system to study data vulnerability; Fuentes Valenzuela and Degleris [14] applied it to a DC-like transport model to learn prices, and honestly report that on real U.S. data the method *could not even overfit a single hour* of EIA-930 observations, let alone generalize.

This gap — a principled, efficient machinery that fails when confronted with real data — is the starting point of this project.

2 Problem

Forward model. Consider a transmission network with n buses and m lines, incidence matrix $A \in \mathbb{R}^{n \times m}$, generation $g \in \mathbb{R}^n$, flows $p \in \mathbb{R}^m$, and net demand $d \in \mathbb{R}^n$. We take the standard DC-OPF formulation as a quadratic program:

$$\begin{aligned} & \underset{g, p}{\text{minimize}} && \frac{1}{2} g^\top \text{diag}(f) g + \frac{\tau}{2} \|p\|_2^2 \\ & \text{subject to} && g - d = Ap, \\ & && 0 \leq g \leq g^{\max}, \quad |p| \leq p^{\max}. \end{aligned} \tag{1}$$

We treat the parameters $\theta = (f, g^{\max}, p^{\max}) \in \mathbb{R}^k$ as unknowns and denote the (single-valued, piecewise-affine) solution map $g^*(\theta, d)$.

Inverse model. Given T observed (demand, generation) pairs $\{(d_t, g_t^{\text{obs}})\}_{t=1}^T$ from EIA-930 and a known topology A , we seek

$$\underset{\theta \in \Theta}{\text{minimize}} \quad \frac{1}{T} \sum_{t=1}^T \ell(g^*(\theta, d_t) - g_t^{\text{obs}}) + \lambda R(\theta), \quad (2)$$

where ℓ is a loss (quadratic or Huber), R is a convex regularizer, and Θ encodes sign and monotonicity constraints (prices non-negative, capacities upper-bounded by nameplate, etc.). Problem (2) is a bilevel program; the forward problem (1) is convex in (g, p) so its KKT conditions admit implicit differentiation, and $\nabla_{\theta} g^*$ can be computed analytically [3, 5]. We solve (2) by projected (sub-)gradient descent, with gradients supplied by `cvxpylayers` [1].

Why the natural formulation fails on real data, and what to fix. Fuentes Valenzuela and Degleris [14] learn only the price vector f against a quadratic residual. On EIA-930 the model cannot fit even one hour; the authors attribute this to either hyperparameter tuning or model misspecification, but leave the question open. I argue the diagnosis is structural, and propose three targeted convex extensions:

1. **Extended parameter vector.** Dispatch in practice is driven as much by binding *capacity* constraints as by costs. Learning only f leaves no freedom to explain why a cheap generator fails to fully dispatch. We extend θ to include $g^{\max}$ and $p^{\max}$, which makes the inverse problem a bilevel program with both cost-parameter and constraint-activation sensitivities [3, 13].
2. **Stratified, time-varying parameters.** Fuel prices and renewable penetration vary by hour-of-day and season, so a single scalar f_i per generator is doomed. We adopt a Laplacian-regularized stratified model [18, 19] over a stratification graph of (hour-of-day $\times$ month), so $f^{(s)}$ varies per stratum s and is regularized to be close between adjacent strata. This stays a convex fitting problem and borrows strength from data-rich strata to data-poor ones.
3. **Robust loss with active-set sparsity.** Real generation data has sensor noise, unit commitment effects, and outliers from derates; we use Huber loss on the residual and add an ℓ_1 penalty on slacks of a lifted formulation to encourage a sparse set of binding constraints consistent with each datum.

Downstream use: sensitivity and counterfactuals. Once $\hat{\theta}$ is learned, standard sensitivity analysis [6] gives the Jacobian $\partial g^*/\partial d$ and marginal emission factors $\nabla_d C = (\partial g^*/\partial d)^\top c$ for any carbon intensity vector c . Counterfactual policy experiments (add 10 GW of solar in a region; impose a \$50/ton carbon tax via a dual perturbation; derate a transmission corridor) become one-line perturbations to $\hat{\theta}$ or d .

3 Data and evaluation

Synthetic validation. I will use PGLib-OPF test cases (14, 30, 57, 118, and 500 buses) [4] with load profiles drawn as in OPFData [15]: 300k solved AC-OPF instances per case. Synthetic ground truth lets me report recovery scores $\cos(\angle(\hat{\theta}, \theta^*))$ as a function of dataset size and noise level, and ablate the three extensions above against the Fuentes Valenzuela and Degleris [14] baseline.

Real data. I will use EIA Form 930 *Hourly Electric Grid Monitor* data [12], accessible via the `gridemissions` package and covering all 65 U.S. balancing authorities at hourly resolution since 2015. For each balancing authority region I have hourly demand, generation by fuel, and inter-BA interchange, which combined with a coarse transmission topology constitutes all inputs required by (1).

Planned figures. (1) *Recovery curves* on IEEE-118 showing $\cos(\angle(\hat{f}, f^*))$ vs. training-set size, with and without each extension, against the Fuentes Valenzuela and Degleris [14] baseline. (2) *Learned merit-order curve* (generator unit costs sorted) overlaid against the published merit-order for a real BA, showing that the method recovers the physically expected ordering (nuclear < coal < gas peaker). (3) *MEF heatmap* over the contiguous U.S., hour-of-day $\times$ season, derived from the stratified learned model. (4) *Counterfactual bar charts*: generation-mix shift under a \$50/ton carbon tax and under a +10 GW solar injection in ERCOT. (5) *Per-hour residual plot* for one BA over one week, contrasting our extended model against the one-price-per-generator baseline of [14].

4 Compute and deliverables

DC-OPF on IEEE-118 is a ~ 300 -variable QP that solves in under 100 ms with OSQP [17]; 500 training examples $\times$ 1000 outer iterations is feasible on a laptop and requires no GPU. Code will be released as a Python package built on CVXPY [10], `cvxpylayers` [1], OSQP, and PowerModels.jl [9] for topology import. The final report will follow the four-figure synthetic-plus-real structure of the course sample reports.

References

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